# 1A ATOMS, ELEMENTS AND MOLECULES

SPECIFICATION REFERENCE 1.1

## LEARNING OBJECTIVES

- Know the terms atom, element, ion, molecule, compound.

## THE PURPOSE OF THIS SECTION

In your previous study of chemistry, you have learned about several terms at an introductory level. In this book, you will study them in greater depth.

This topic is designed to revise some of the terms that you have met before, and will meet again in more detail.

## WHAT IS AN ELEMENT?

We can start by looking at a simplified version of the Periodic Table of Elements. You have already learned something about the Periodic Table and will learn more about it in **Topic 2**.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | H | | | | He |
| Li | Be | B | C | N | O | F | Ne |
| Na | Mg | Al | Si | P | S | Cl | Ar |

Each box in the Periodic Table contains a capital letter, or more often a pair of letters. The first one is always a capital letter and the second one is never a capital letter. Each letter or pair of letters represents an element. You will know some of them from your everyday life. For example, oxygen is an element you breathe, and iron is an element used to make bodies of cars and bicycles.

You may have been taught that an element is a substance that contains atoms of only one type. Elements are chemically the simplest substances so they cannot be broken down using chemical reactions. For example, neon is an element because it contains only neon atoms, which have the symbol Ne. It cannot be broken down into atoms of any other element. Water is not an element because it can be broken down into the elements hydrogen and oxygen.

This way of describing an element is sufficient for this stage of your studies, but more details will be required as you progress through this course. For example, some elements contain isotopes. You will learn about isotopes in **Topic 2**. Neon contains three stable isotopes, $^{20}Ne$, $^{21}Ne$, and $^{22}Ne$. All isotopes of the same element have the same number of protons and electrons, but different numbers of neutrons.

## WHAT IS AN ATOM?

You already know what an atom is, because we have used the term to help you understand what an element is. Atoms are far too small to be seen with the human eye. You can see a tiny grain of sand and this contains many billions of atoms of silicon and oxygen. An atom can be described as the smallest part of an element that has the properties of that element. This meaning is fine when you work through **Topic 1**. However, in **Topic 2** you will need to recall that atoms contain even smaller particles (protons, neutrons and electrons).

## WHAT IS A MOLECULE?

This is easier now that we know the meanings of element and atom.

You could describe a molecule as a particle made of two or more atoms bonded together.

If a molecule contains atoms of the same element, then the result is a molecule of an element. For example, a molecule that contains two atoms of hydrogen joined together can be represented by the formula $H_2$. This formula contains only one symbol (H), so it is the formula of an element.

If a molecule contains atoms of two or more different elements, then the result is a molecule of a compound. For example, a molecule that contains two atoms of hydrogen and one atom of oxygen joined together can be represented by the formula $H_2O$. This formula contains two different symbols, so it is the formula of a compound (water).

## WHAT IS A COMPOUND?

Now we can describe a compound. This is a substance containing atoms of different elements combined together.

Note that some compounds contain large numbers of atoms bonded together, but other compounds contain molecules with only two atoms. Some compounds contain oppositely charged ions. We will explore these differences in later sections of this book.

## WHAT IS AN ION?

One way to describe an ion is as a species consisting of one or more atoms joined together and having a positive or negative charge.

Note that this is not a description of how an ion is formed. **Topic 3A.1** will cover the formation of ions.

An ion with a positive charge is called a cation. An ion with a negative charge is called an anion.

**Table A** shows diagrams to illustrate the terms referred to. Each atom is shown as a circle containing the symbol of an element. The lines show bonds between atoms.

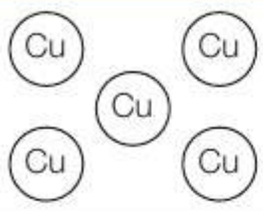
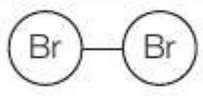
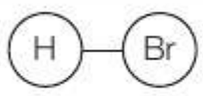
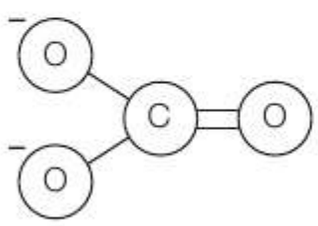
| TERM | DIAGRAM | NAME | SYMBOL OR FORMULA | NOTE |
|---|---|---|---|---|
| element |   | copper | Cu | This is an element. All the atoms are the same. |
| atom |   | helium | He | This is an atom of an element. |
| molecule |   | bromine | $Br_2$ | This is a molecule of an element. The atoms are the same. |
| compound |   | hydrogen bromide | HBr | This is a molecule of a compound. The atoms are different. |
| ion |   | carbonate | $CO_3^{2-}$ | This is an ion. There are two negative charges shown. |

**table A** Illustrations of terms used in this topic.

## OTHER TERMS

Elements that are made up of single atoms are described as monatomic. One example is helium, the gas used in weather balloons. The symbol for helium is He.

Elements and compounds made up of two atoms joined together are described as diatomic. The two main diatomic gases in the atmosphere are nitrogen ($N_2$) and oxygen ($O_2$).

Elements and compounds with molecules made up of several atoms joined together are described as polyatomic. Examples of polyatomic molecules are phosphorus ($P_4$) and methane ($CH_4$).

The same terms can be used for ions. Chloride ($Cl^-$) is an example of a monatomic ion. Hydroxide ($HO^-$ or $OH^-$) is a diatomic ion. A sulfate ion ($SO_4^{2-}$) is polyatomic as it contains five atoms.

### LEARNING TIP

It is often helpful to refer to a substance by its symbol or formula as well as its name so that your meaning is clear. For example, the word hydrogen could refer to a hydrogen atom (H), a hydrogen molecule ($H_2$) or a hydrogen ion ($H^+$).

### CHECKPOINT

**1.** Classify each of these symbols and formulae as atoms, molecules or ions.

Ne $CO_2$ $H^+$ $S_8$ $Al^{3+}$

**2.** Which of these formulae represent elements, compounds, or neither elements nor compounds? Explain your answer.

$Br_2$ $H_2O_2$ $NO_3$ $O_3$ CaO

### DID YOU KNOW?

The names and symbols of many elements come from the Greek and Latin languages. For example, hydrogen comes from the Greek words for *water* and *producer*. This is because when hydrogen gas is burned, it forms (or 'produces') water.
Another example is copper (Cu) – the symbol Cu comes from the Latin word *cuprum*, which means 'metal from Cyprus'. Cyprus is the island where the Romans obtained much of their copper.